Common LLM Model Formats and Understanding GGUF – Summary Paper

Introduction

Large Language Models (LLMs) are distributed in different model formats depending on their training framework, deployment environment, hardware requirements, and optimization methods. Choosing the correct format is important for efficient AI deployment.

1. Common Model Formats

PyTorch (.pt / .bin / .pth)

PyTorch is one of the most common formats used for training and research. It stores model weights and is widely supported by machine learning frameworks.
Use: Model development, fine-tuning, and research environments.

Safetensors (.safetensors)

Safetensors is a safer and faster alternative to traditional weight files. It avoids security risks associated with serialized objects and provides efficient loading.
Use: Modern LLM distribution, Hugging Face models, and production deployment.

ONNX (.onnx)

ONNX (Open Neural Network Exchange) is a portable format that allows models to run across different AI frameworks and hardware platforms.
Use: Enterprise deployment, edge devices, and optimized inference.

TensorFlow SavedModel

A format developed for TensorFlow-based machine learning applications.
Use: Google Cloud, TensorFlow serving, and production ML systems.

GGUF (.gguf)

GGUF (GPT-Generated Unified Format) is a model file format designed mainly for efficient local running of Large Language Models. It is the successor to the older GGML format and is widely used with llama.cpp-based applications.

Advantages of GGUF:

- Supports model quantization to reduce memory usage.
- Allows LLMs to run on laptops, desktops, and edge devices.
- Provides faster loading and efficient inference.
- Supports different precision levels such as 4-bit, 5-bit, and 8-bit quantization.

Example:

A 70B parameter LLM may require hundreds of GBs in full precision, but a quantized GGUF version can be significantly smaller and run on consumer hardware with acceptable performance.

GGUF vs Other Formats

GGUF is mainly optimized for inference, especially local AI applications. Safetensors and PyTorch formats are preferred for training and fine-tuning, while GGUF is preferred for deploying and running optimized models.

Recommendation

For AI development and fine-tuning: use PyTorch or Safetensors.
For cloud and enterprise deployment: consider ONNX or optimized serving formats.
For local AI assistants and edge deployment: use GGUF with tools such as llama.cpp, Ollama, or similar inference platforms.

Conclusion

Different model formats serve different purposes. GGUF has become an important format for efficient LLM inference because it enables powerful models to run on smaller hardware with reduced memory requirements.